

ENAUTO Lesson 9

Modules, Controllers, and Dynamic Data

Objective

Understand how modules, controllers, functions, and returned values work together in a real automation project.

Core Concepts

- A function is a reusable tool.
- Referencing a function is different from executing it.
- Parentheses execute a function.
- A function can return a value.
- The controller coordinates work while modules perform the work.

Architecture

```
controller.py
|
+--> lesson8.py
      |
      +--> get_hostname()
      +--> get_ip_address()
      +--> get_available_ram()
      +--> check_ram()
      +--> main()
```

Correct Controller

```
import lesson8

available_ram = lesson8.get_available_ram()

print("Available RAM:", available_ram, "GiB")

lesson8.check_ram(available_ram)
```

What We Learned

lesson8.get_available_ram -> the function
lesson8.get_available_ram() -> run the function
available_ram = lesson8.get_available_ram() -> run it and save the returned value.

Debugging Lessons

- Missing function arguments produce TypeError.
- Undefined variables produce NameError.
- Import modules without '.py'.
- Printing a function prints the function object.
- Printing the returned value requires calling the function first.

Automation Mindset

Today's code is the same design used later with Netmiko, RESTCONF, and APIs:

```
device = connect_to_switch()
facts = gather_facts(device)
backup = get_running_config(device)
```

```
save_config(backup)
```

Only the function names change; the architecture stays the same.

Takeaway

The most important lesson today was understanding the difference between a function, executing a function, and storing the value that the function returns.